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Eco Park: A Low-Cost and Sustainable Approach for Smart Parking Systems

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***Abstract*- The challenges and opportunities related with the management and optimisation of parking lots in urban areas are addressed in this research paper. Parking lots have grown increasingly obsolete, necessitating tremendous personnel to manage and maintain. These lots frequently lack user-friendly features, such as real-time availability information, resulting in motorist irritation and wasted time. Furthermore, the increasing size of luxury vehicles and the scarcity of parking places in metropolitan areas complicate parking management.To address these concerns, we provide a ground-breaking solution based on the Internet of Things (IoT). To deliver real-time parking availability data, reduce congestion, and improve the entire parking experience for users, our suggested system design combines sensors, gateways, a cloud-based platform, and a mobile application. Our research emphasises the importance of IoT in changing traditional parking systems into intelligent and efficient solutions. We provide a thorough explanation of the system design and its components, backed up by a comparative analysis that demonstrates the efficacy of our method in contrast to existing systems..By leveraging IoT technologies, our research contributes to the evolution of parking management solutions. Our insights will be useful to policymakers, urban planners, and parking facility operators in establishing sustainable, cost-effective, and user-centric parking solutions. Urban regions can improve their overall quality of life by implementing our results and providing efficient and easy parking experiences.**

Keywords- Cost-effective solutions, Eco-friendly parking technologies, EcoPark, Energy-efficient parking systems, Environmental sustainability, Green infrastructure for parking, Green parking, Intelligent parking management, IoT-based parking, Low-cost, Optimization of parking resources, Smart city initiatives, Smart parking systems, Sustainable, Sustainable urban mobility.

I. INTRODUCTION

The notion of the Internet of Things (IoT) originated with gadgets that have identification and communication devices. Remote computers with Internet access can track, control, and monitor the devices. IoT enhances Internet usage by facilitating communication and, as a result, inter-networking of devices and physical things, or "Things." The two key terms in IoT are "internet" and "things." The phrase "Internet" refers to a vast global network of interconnected servers, laptops, tablets, and mobile devices that use universally approved protocols and connecting systems. You may use the Internet to send, receive, and communicate information.As s

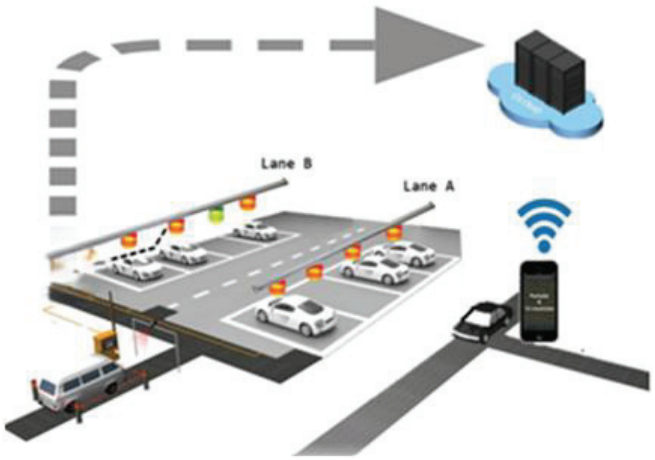


Fig. 1. Parking System [1]

In English, the word thing has several different meanings and applications. A 'Thing' is a term used to refer to a physical thing, an action or notion, a situation or activity, according to the dictionary definition. In general, IoT is a network of interconnected devices and physical goods, with a high number of gadgets capable of gathering data from a distance. It envisions a future in which objects (wearables, watches, alarm clocks, home gadgets, and surrounding goods) become intelligent and alive by sensing, processing, and communicating via implanted small devices that interact with remote objects or people via connectivity. Because of the scalability and reliability of cloud computing, developers may create and host programmes on it. Because it serves as a platform where all sensor data can be stored and accessed from remote places, the cloud is a perfect partner for IoT.Because it serves as a platform where all sensor data can be stored and accessed from remote places, the cloud is a perfect partner for IoT.Because of these factors, the merger of both technologies led in the invention of a new technology known as Cloud of Things (CoT). Things (nodes) in CoT may be accessed, monitored, and controlled from any distant location through the cloud. Because of the cloud's incredible scalability, any number of nodes may be added or removed from the IoT system in real time. IoT may be characterised in the following ways:

Physical Item + Controller, Sensor, and Actuators + Internet = Internet of Things

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With the rise of the Internet of Things, the dream of building a Smart City is becoming a reality. Car parking facilities and traffic management systems are two main challenges that smart cities address.

II. NEED FOR IOT CLOUD AND DATABASE INTEGRATION

The integration of a database is a critical component of an IoT-based parking system. The storage capacity required to handle the massive volumes of data created by IoT devices, such as parking spot information and user data, necessitates the deployment of an efficient and dependable database system. MySQL, a popular and dependable open-source relational database management system, is a popular choice for this purpose.

The parking system can store and manage parking spot information, user profiles, reservations, and other important data by including a database, such as MySQL, into the overall architecture. With its scalable and on-demand storage capacity, the Cloud is a great environment for storing the MySQL database, guaranteeing effective data management and retrieval.

The integration of a database enables numerous parking system functionalities. Users can view real-time parking availability, make bookings, and get historical statistics. IoT devices, such as cameras or sensors, collect and transmit data to the Cloud, where it is stored in the MySQL database. The database can be accessed by the Cloud-based application or user interface to retrieve and show pertinent information to users.

A comprehensive and scalable solution for an IoT-based parking system is created by combining Cloud computing, IoT, and a database such as MySQL. The Cloud provides infrastructure, storage, and processing power, while IoT devices collect real-time data and the database stores and manages it efficiently. These components work together to provide seamless data integration, real-time monitoring, effective resource allocation, and data-driven decision-making.Cloud computing, IoT, and a database like MySQL provides various benefits for a parking system, including efficient data storage, real-time monitoring, enhanced data processing, and scalability. This integration guarantees a dependable and scalable solution for managing parking data and providing a consistent user experience.

III. LITERATURE REVIEW

The author[1] Describe the architecture of their proposed system, which comprises sensors, gateways, a cloud-based platform, and a mobile application. They demonstrate how these components interact to provide real-time parking availability, ease congestion, and improve the overall user experience. The paper provides a full evaluation of the current state of parking systems and proposes a novel IoT-based solution to address their limitations. To demonstrate the effectiveness of their approach, the authors provide detailed explanations of the system design and its components, as well as comparisons to comparable current systems.

The research of Ajam and Dholay [2] focuses on developing a smart parking system that can find the best parking spot for a car by utilising numerous sensors and machine learning algorithms. The authors investigate the current state of smart parking systems, as well as their benefits and drawbacks. The paper provides an in-depth assessment of the present status of smart parking systems and proposes a novel technique for locating the best parking spot for a car by combining numerous sensors and machine learning algorithms. The authors compare their proposed system to existing options and demonstrate that it outperforms them in terms of accuracy, efficiency, and cost-effectiveness.

Thangam et al. [3] propose a smart parking reservation system based on Internet of Things (IoT) technologies and the Raspberry Pi hardware platform. The authors look at the current state of parking systems, the challenges they confront, and the potential benefits of a smart parking reservation system. The study examines the current state of parking systems in depth and proposes a novel smart parking reservation system that takes use of IoT technology and Raspberry Pi devices. The authors describe the system design and the benefits of using Raspberry Pi devices before comparing their proposed system to various current alternatives, demonstrating its superiority in terms of cost-effectiveness, scalability, and usability.

Kanteti et al. [4] propose an intelligent parking system for urban business zones. The authors investigate the current state of commercial parking systems, their shortcomings, and the potential benefits of a smart parking system. The paper examines the current state of parking systems in depth and proposes a novel smart parking system for urban commercial districts. The authors describe the system architecture and the benefits of using a smart parking system for commercial areas, and they compare their proposed system to other current systems, demonstrating its superiority in terms of scalability, usability, and revenue generation potential.

Vakula and Kolli[5] propose a low-cost smart parking solution for smart cities. The authors investigate the current state of parking systems, their limitations, and the possible benefits of a smart parking system for smart cities. The report examines current parking systems in depth and proposes a low-cost smart parking solution for smart cities. The authors describe the system architecture and benefits of using a low-cost smart parking system for smart cities, as well as compare their proposed system to other existing solutions, demonstrating its superiority in terms of cost-effectiveness and ease of deployment.

Elsonbaty and Shams[6] offer a smart parking management system. The authors investigate existing parking systems, their flaws, and the potential benefits of a smart parking management system. The article describes the system architecture, which includes wireless sensors, a cloud-based platform, and a mobile application. The authors explain how wireless sensors identify the presence of a vehicle and send data to a cloud-based platform. The data is processed via a cloud-based platform, which provides real-time parking availability information and simplifies the payment process. Users can browse for available parking spaces, reserve them, and pay for them using the smartphone app. Relevant research in the field of smart parking systems, highlighting their advantages and disadvantages.

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Agarwal et al. [7] discuss a ground-breaking smart parking system based on Internet of Things (IoT) technology. The authors investigate existing parking systems, their flaws, and the potential benefits of a smart parking system. The authors explain how ultrasonic sensors identify cars and send data to the Raspberry Pi. The data is processed by the Raspberry Pi, which provides real-time parking availability information. The smartphone application allows users to locate and reserve available parking places. The authors describe the system design and the benefits of using a smart parking system, as well as compare their proposed system to other current solutions, demonstrating its superiority in terms of cost-effectiveness and ease of installation.

Hanif et al. [8] propose a smart parking reservation system based on SMS. The authors investigate existing parking systems, their flaws, and the potential benefits of a smart parking reservation system. The authors discuss how the system allows users to reserve parking spaces by using SMS technology. The server processes reservations and provides real-time parking availability information. Furthermore, the authors propose a pricing strategy based on demand and supply. The study examines current parking systems in depth and offers a novel smart parking reservation system based on SMS technology. The authors describe the system architecture and the benefits of using a smart parking reservation system, as well as compare their proposed system to other current alternatives, demonstrating its superiority in terms of simplicity and efficiency.

Farooqi et al. [9] describe the development and implementation of Parking, a smart parking management system. The authors investigate existing parking systems, their flaws, and the potential benefits of a smart parking management system. The system architecture is made up of IoT sensors, a server, and a mobile app. The Internet of Things sensors identify the presence of autos in parking lots and relay the information to a server for processing. The smartphone software displays real-time parking availability and allows users to book and pay for parking. The authors describe the system architecture and the benefits of using a smart parking management system, as well as compare their proposed system to other current systems, demonstrating its superiority in terms of accuracy and real-time data.

Hainalkar and Vanjale[10]demonstrate a smart parking system including pre- and post-reservation, charging, and a traffic app. The authors present a review of the literature on smart parking systems, emphasising their limitations and the necessity for a more comprehensive system that incorporates elements like as reservation and invoicing. The study provides an in-depth examination of smart parking systems and suggests a revolutionary solution that incorporates pre and post reservation, invoicing, and a traffic app. The authors describe the difficulties in constructing such a system and provide solutions to these difficulties. They compare their proposed system to existing alternatives and emphasize its merits, offering insights into the possible advantages of establishing a smart parking system with reservation and invoicing capabilities.

Ramasamy, Manickam, and colleagues [11] investigate the design and implementation of an IoT-based smart parking system for big parking lots. The writers address the difficulties in managing parking spaces and highlight the potential of IoT technology to revolutionise the parking industry. They propose a system architecture that makes use of sensors, wireless connectivity, and data analytics to provide real-time monitoring of parking availability, effective space management, and user convenience. The review comprises an assessment of the proposed system's benefits, experimental results, and implementation and scalability considerations. Overall, the study emphasises the benefits of IoT technology in improving parking management and the overall parking experience in metropolitan locations.

Gupta, Aniket, et al. [12] leverage IoT technologies to develop a smart car parking management system. The authors analyse the growing challenges of urban parking management and the possibility of IoT in addressing these issues. To construct an intelligent parking system, they offer a system architecture that incorporates numerous IoT components such as sensors, actuators, and communication technologies. The benefits of the proposed system are highlighted in the review, which include real-time parking space availability, effective parking spot allocation, and improved customer convenience. The paper also analyses the IoT-based parking management system's possible uses and gives experimental findings to prove its effectiveness.

The literature evaluations on smart parking systems revealed many shortcomings or limitations in the studies. Some of the most common limitations observed include a lack of comprehensive information about potential implementation challenges, a lack of thorough discussion about the drawbacks or limitations of the proposed systems, an inadequate consideration of security concerns associated with IoT-based systems, and a lack of emphasis on user acceptance and behaviour change required for successful system adoption. Scalability constraints, privacy concerns, sensor data reliability and accuracy, challenges with SMS-based reservation systems, and potential limitations in terms of system integration and infrastructure were all recognised as areas requiring additional consideration. It should be noted that the significance and relevance of these constraints may differ based on the specific environment and requirements of any smart parking system under discussion.

IV. PROPOSED WORK

A. Algorithm:

Step 1: Go to the website for the Car Parking Reservation.

Step 2: Enter the Name

Step 3: Enter the Car Number

Step 4: Choose the date when you need to reserve a parking space.

Step 5: To book, choose a Start Time and End Time.

Step 6: If it is available, the website will prompt you for your reservation.

Step 7: The automobile must arrive in the parking place within a certain amount of time before the verification process is completed.

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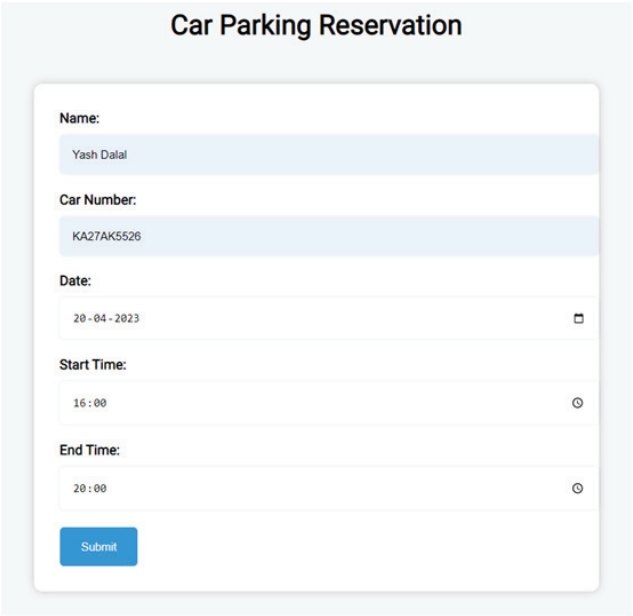


Fig. 2. Car Parking Reservation Schedule

The following flow chart explains the whole process of booking a parking space, parking a car in that slot, and departing the parking area.

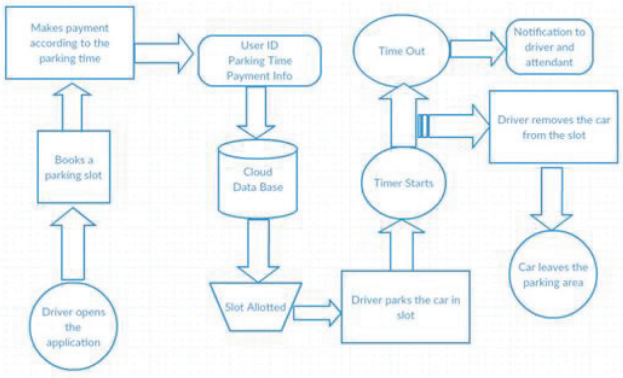


Fig. 3. Flow Chart of the System [1]

To get real-time parking status of parking slots, the suggested system employs an Ultrasonic Sensor to identify the presence of an automobile in a specific parking slot. Each parking place in each car park incorporates an Ultrasonic Sensor, which will be linked to the Local Control Unit. The

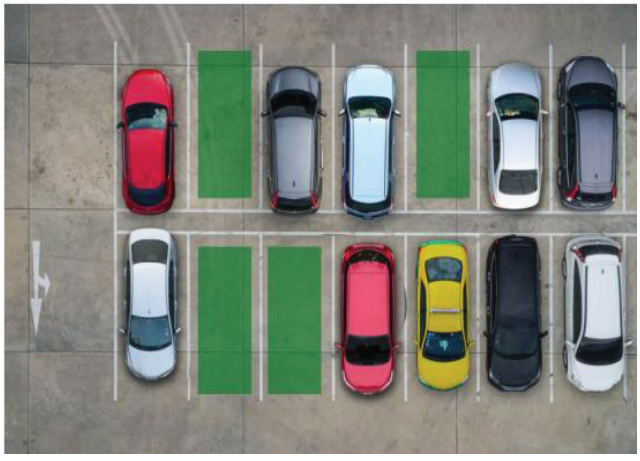


Fig. 4. Parking Status Update Process

information given by each sensor will be used to identify the overall number of cars parked in the parking space. The parking spot information is shared by the local control unit with the Web API on the cloud server. Figure 4 depicts the procedure of estimating the total number of available parking places.

B. System Overview:

The suggested system takes a different approach, with three important modules: the user module, the parking space module, and the dynamic algorithm. This configuration includes a strategy for minimizing the time required for parking spot identification. When a user selects a parking spot, the information associated with that location is displayed to the user.

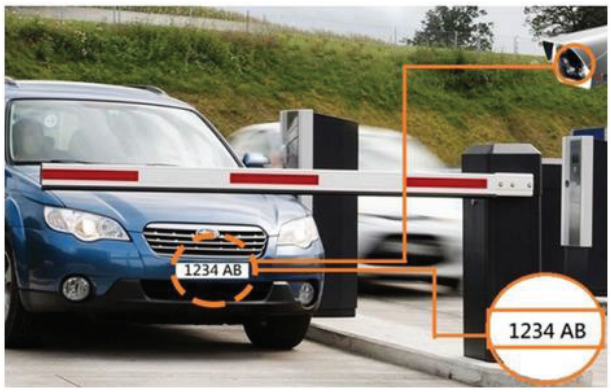


Fig. 5. System Overview Checking Process

During the booking process, the parking space's status is displayed as "Partially Reserved" until the reservation is finished. When a user successfully reserves a parking place, it is marked as "Reserved" for the term given. The ultrasonic sensor continuously monitors each slot to detect the presence of automobiles in unreserved parking spaces If a vehicle is not detected in a specific location, it is labelled "Available." The status of each parking place is updated in real-time, guaranteeing that users have synchronized and correct information. The system provides a cost-effective and efficient solution for parking management by eliminating the need for a camera-based system and immediately updating the database via the suggested IoT configuration. This method optimizes the process of locating available parking spaces, improves user experience, and allows for real-time parking system updates.

C. System Architecture:

The suggested system is built on a cloud-computing environment for processing raw data, a Web API Application for data processing, a Mobile Application for clients to interface with the system, and a database for storing parking spot information and user data. The Arduino module positioned at each parking space will send the parking slot information to the cloud server; the application server will process this information and communicate it with the Application or User Interface. On the application, users will be able to monitor the real-time availability status of each parking place. The suggested system's architecture is as follows shown in the figure 6.

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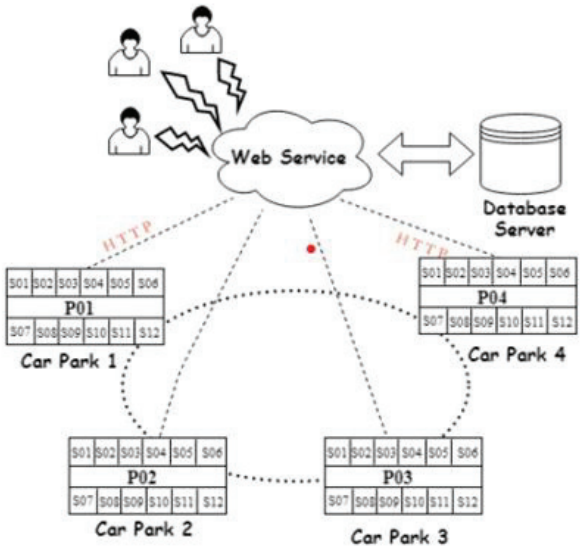


Fig. 6. Architecture of the System [2]

1) Local Unit: The Web API will serve as the interface between the database and applications such as the Arduino module and user application on the cloud-based server. The Arduino module shares parking spot information, which the Web API stores in the database. The Web API also delivers



Fig. 7. Working of Model

parking spot information to the user application depending on the location given by the user.

2) Cloud-Based Server: The cloud-based server is made up of the Web API, which serves as an interface between the database and applications such as the Arduino module and the user application. The Arduino module shares parking spot information, which the Web API stores in the database. The Web API also delivers parking spot information to the user application depending on the location given by the user.

3) Client Software: The client software is an Android app that connects to the Web API over an Internet connection. Using this application, users may acquire information about nearby parking lots based on their present position or search for parking lots at a certain area at a specific time.

The table 1 shows the number of pre-booked slots and new excess requests for specific hours, as well as the percentage of pre-booked slots that are occupied. The table is described below:

Time: The particular time slots expressed in a 12-hour format (for example, 10:00 a.m., 11:00 a.m., 12:00 p.m., and so on).

TABLE I. EFFICIENT TIME ALLOCATION [4]

Time	*Pre-Booked slots at Particular Time*	*New excess requests for the time & which get vacated at that time*	*Occupancy of Pre-booked slots (%)*
10:00 am	10	7	70
11:00 am	15	8	53.33
12:00 pm	12	8	66.67
1:00 pm	15	8	53.33
3:00 pm	11	5	45.45
5:00 pm	20	9	45
7:00 pm	25	10	64

Pre-Booked Slots at a Specific Time: The amount of slots reserved or pre-booked in advance for each specified time.

New surplus requests for the time and those that are vacated at the time: The number of new excess requests received for that time slot, as well as the number of slots that become available (vacated) owing to vehicles leaving.

Pre-booked Slot Occupancy (%): The percentage of pre-booked slots that are occupied at a given moment, computed by dividing the number of occupied pre-booked slots by the total number of pre-booked slots and multiplying by 100.

This data provides insight into the utilization and occupancy patterns of pre-booked slots at various times of the day. It can be useful for analyzing parking demand, optimizing slot distribution, and determining the overall system efficiency.

V. CONCLUSION

Finally, this study suggests an IoT-based solution to the issues of managing and optimizing parking lots in metropolitan settings. Our system architecture provides real-time parking availability data, congestion reduction, and better user convenience by integrating sensors, gateways, a cloud-based platform, and a mobile application. Our solution comprises three components: the backend, frontend, and website/mobile app. By continuously refreshing camera data and achieving high precision in parking availability monitoring (99.7% accuracy), our technology significantly reduces parking search time. Users can conveniently book parking spaces in advance through the mobile app and website, both of which actively monitor parking availability via cameras. The system is customizable to meet specific requirements and includes essential features like reporting and integrated payment support. Overall, our solution offers a high quality and high-performance parking management system that maximizes parking spot utilization and enhances the user experience.

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REFERENCES

[1] Khanna, Abhirup, and Rishi Anand. "IoT based smart parking system." 2016 international conference on internet of things and applications (IOTA). IEEE, 2016.

[2] Zajam, Ajay, and Surekha Dholay. "Detecting efficient parking space using smart parking." 2018 9th International Conference on Computing, Communication and Networking Technologies (ICCCNT). IEEE, 2018.

[3] Thangam, E. Cassin, et al. "Internet of Things (IoT) based smart parking reservation system using raspberry-pi." International Journal of Applied Engineering Research 13.8 (2018): 5759-5765.

[4] Kanteti, Dharmini, D. V. S. Srikar, and T. K. Ramesh. "Smart parking system for commercial stretch in cities." 2017 International Conference on Communication and Signal Processing (ICCSP). IEEE, 2017.

[5] Vakula, D., and Yeshwanth Krishna Kolli. "Low cost smart parking system for smart cities." 2017 International Conference on Intelligent Sustainable Systems (ICISS). IEEE, 2017.

[6] Elsonbaty, Amira, and Mahmoud Shams. "The smart parking management system." arXiv preprint arXiv:2009.13443 (2020).

[7] Agarwal, Yash, et al. "IoT based smart parking system." 2021 5th international conference on intelligent computing and control systems (ICICCS). IEEE, 2021.

[8] Hanif, Noor Hazrin Hany Mohamad, Mohd Hafiz Badiozaman, and Hanita Daud. "Smart parking reservation system using short message services (SMS)." 2010 International Conference on Intelligent and Advanced Systems. IEEE, 2010.

[9] Farooqi, Norah, et al. "UParking: Developing a smart parking management system using the internet of things." 2019 Sixth HCT Information Technology Trends (ITT). IEEE, 2019.

[10] Hainalkar, Gayatri N., and Mousami S. Vanjale. "Smart parking system with pre & post reservation, billing and traffic app." 2017 International Conference on Intelligent Computing and Control Systems (ICICCS). IEEE, 2017.

[11] Ramasamy, Manickam, et al. "IoT based smart parking system for large parking lot." 2018 IEEE 4th International Symposium in Robotics and Manufacturing Automation (ROMA). IEEE, 2018.

[12] Gupta, Aniket, et al. "Smart car parking management system using IoT." Am. J. Sci. Eng. Technol 2.4 (2017): 112-119.

[13] Ismail, Mohd Mustari Syafiq, et al. "IoT Based Smart Parking System." Journal of Physics: Conference Series. Vol. 1424. No. 1. IOP Publishing, 2019.

[14] Sarangi, Manisha, Santos Kumar Das, and Korra Sathya Babu. "Smart parking system: survey on sensors, technologies and applications." 2019 1st International Conference on Advances in Information Technology (ICAIT). IEEE, 2019.

[15] Kadhim, Maher Hassan. "Arduino Smart Parking Manage System based on Ultrasonic." 2018, Int. Journal of Engineering & Technology 7.3.20 (2018): 494-501.

[16] Islam, Mohammed Rezwanul, et al. "Smart parking management system to reduce congestion in urban area." 2020 2nd international conference on electrical, control and instrumentation engineering (ICECIE). IEEE, 2020.

[17] Zulfiqar, Haidar, et al. "A survey on smart parking systems in urban cities." Concurrency and Computation: Practice and Experience (2021): e6511.

[18] Mishra, Brundaban, et al. "Smart parking system." SYSTEM 5.04 (2018).

[19] Hasan, Md Omar, Md Motaharul Islam, and Yazed Alsaawy. "Smart parking model based on internet of things (IoT) and TensorFlow." 2019 7th International Conference on Smart Computing & Communications (ICSCC). IEEE, 2019.

[20] Pham, Thanh Nam, et al. "A cloud-based smart-parking system based on Internet-of-Things technologies." IEEE access 3 (2015): 1581-1591.

[21] Bachhav, Jayakshei Dadaji, and M. A. Mechkul. "Smart car parking system." International Research Journal of Engineering and Technology (IRJET 4.06 (2017).

[22] Zajam, Ajay, and Surekha Dholay. "Detecting efficient parking space using smart parking." 2018 9th International Conference on Computing, Communication and Networking Technologies (ICCCNT). IEEE, 2018.

[23] Sarangi, Manisha, et al. "IoT aware automatic smart parking system for smart city." Cognitive Informatics and Soft Computing: Proceeding of CISC 2019. Springer Singapore, 2020.

[24] Tiwari, Rajeev, et al. "Automated Parking System-Cloud and IoT based." Int. J. Eng. Adv. Technol 8 (2019): 116-123.

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